

- [CPSC 375: High-Performance Computing](#)

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Spring 2026

- [Syllabus](#)
- [Schedule](#)
- [Upload](#)
- [The Cluster Project](#)

Homework 21

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. What's wrong with the following program fragment?

```
if (my_rank == 0) {
    MPI_Reduce(&x, &y, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);
} else {
    MPI_Recv(&y, 1, MPI_INT, 0, 0, MPI_COMM_WORLD, MPI_STATUS_IGNORE);
}
```

Correct the errors.

2. Given two consecutive calls to MPI_Reduce:

```
MPI_Reduce(&a, &b, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);
MPI_Reduce(&c, &d, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);
```

and the values:

Process 0: a=1, c=2

Process 1: a=2, c=1

Process 2: a=1, c=2,

determine the values stored in b and d on process 0 and explain why the result depends on call order rather than variable names.

3. What is the difference between MPI_Reduce and MPI_Allreduce? Describe a scenario where MPI_Allreduce is necessary.
4. For an array of size 16 distributed across 4 processes, describe the index assignments for block partitioning, cyclic partitioning, and block-cyclic partitioning with block size 2. Which strategy is more appropriate when work per element is uniform versus when it varies significantly? Justify your answer.
5. Assume $n = 8$, $p = 4$, and process 0 initially holds $A = [0,1,2,3,4,5,6,7]$. Determine what each process receives after applying MPI_Scatter. If each process multiplies its elements by 2 and calls MPI_Gather, what is the final array on process 0?

6. What is wrong with the following program?

```
double local_val, result;
MPI_Reduce(&local_val, &result, 1, MPI_DOUBLE, MPI_SUM, 1, MPI_COMM_WORLD);

if (my_rank == 0) {
    printf("Sum = %f\n", result);
}
```

Correct it so it produces the intended result.

7. Explain why the following code is invalid and describe what could happen at runtime.

```
if (my_rank == 0)
    MPI_Bcast(&x, 1, MPI_INT, 0, MPI_COMM_WORLD);
else
    MPI_Bcast(&x, 2, MPI_INT, 0, MPI_COMM_WORLD);
```

- **Welcome: Sean**

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